User Case Diagram & SDLC Selection

Course title: Software Engineering Laboratory
Course code: CSE-3206
Session: July-December 2026

Date of Submission: *27 August, 2026*

Submitted to-

Rumana Yasmin
Department of Computer Science and Engineering
Bangladesh University of Professionals

Tanjim Taharat Aurpa
Department of Data Science and Engineering
University of Frontier Technology, Bangladesh

ID	Name
24524203078	ZAIMA NAHIYAN MUSTAFA
24524203082	MAHMUDA
24524203134	ISRAT JAHAN

1. Problem Analysis

FixMyBD is a centralized platform for reporting and tracking public issues in Bangladesh. Citizens can report issues, add location and evidence, track progress and give feedback. Authorities can review reports, respond to citizens, update status, and add progress information. The system administrator manages users, authorities, categories, locations, reports and system monitoring.

The main problem is that public issue reporting and follow-up can be difficult. Reports can be scattered across different channels, and citizens may not know what happens after a report is submitted. FixMyBD brings the information into one structured system.

2. Use Case Diagram

In FixMyBD, the main actors are Citizen, Authority/Department and System Administrator. The system boundary contains the functions provided by FixMyBD.

2.1 Actors of FixMyBD

- Citizen - registers and logs in, reports public issues, adds location and evidence, tracks reports, supports issues, comments, and gives feedback.
- Authority / Department - views reports, reviews and assesses issues, responds to citizens, updates status, adds progress information, uploads evidence, and responds to feedback.
- System Administrator - manages users, authorities, categories, locations, reports, moderation, analytics and system notifications.

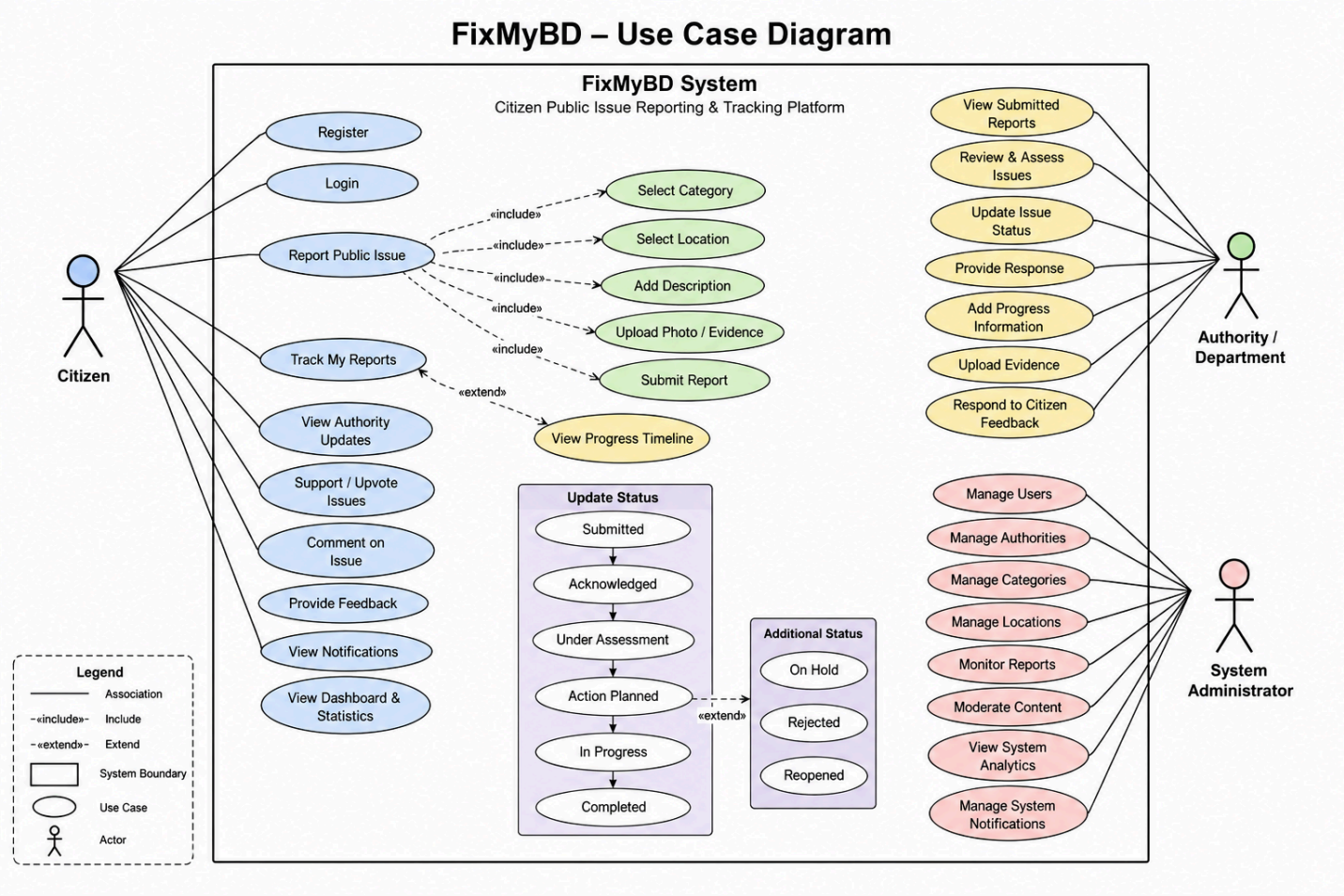
2.2 Main Use Cases

Citizen: Register / Login; Report Public Issue; Select Category; Select Location; Add Description & Photo; Submit Report; Track Report; View Authority Updates; Support / Upvote Issue; Comment; Provide Feedback; View Notifications.

Authority / Department: View Reports; Review Submitted Issues; Assess / Verify Information; Respond to Citizens; Update Issue Status; Add Progress Information; Upload Supporting Evidence; Respond to Citizen Feedback; Mark Completed.

System Administrator: Manage Users; Manage Authorities; Manage Categories; Manage Locations; Monitor Reports; Moderate Inappropriate Reports; View Dashboard & Analytics; Manage System Notifications.

2.3 Use Case Diagram Layout



2.4 Important Relationships

- Citizen interacts with Register / Login, Report Public Issue, Track Report, Support / Upvote Issue, Comment and Provide Feedback.
- Reporting can be shown as a group of related functions: Select Category, Select Location, Add Description & Photo and Submit Report.
- Authority / Department interacts with View Reports, Review Submitted Issues, Assess / Verify Information, Respond to Citizens, Update Issue Status, Add Progress Information and Mark Completed.
- System Administrator interacts with all management and monitoring functions.
- The source project defines the authority status flow as Submitted → Acknowledged → Under Assessment → Action Planned → In Progress → Completed → Citizen Feedback.
- Additional report statuses are On Hold, Rejected and Reopened.

3. Software Development Life Cycle (SDLC) Selection:

Table 1: Comparison Matrix with Different SDLC Models

Priority	Criteria	Waterfall	V-Shape	Iterative	Spiral	Agile	Prototype
5	Changing requirements	No	No	Yes	Yes	Yes	Yes
5	Continuous user feedback	No	No	Yes	Yes	Yes	Yes
5	Incremental delivery	No	No	Yes	Yes	Yes	Yes
4	Modular development	Yes	Yes	Yes	Yes	Yes	No
3	Rapid delivery	No	No	Yes	Yes	Yes	Yes
3	Risk analysis	No	No	No	Yes	No	No
5	Easy adaptation after testing	No	No	Yes	Yes	Yes	Yes
Total 30	**Overall**	**4**	**4**	**27**	**30**	**30**	**22**

Justification

The results are as follows: Waterfall (4), V-Shaped (4), Iterative (27), Spiral (30), Agile (30) and Prototype (22).

The Agile and Spiral models achieved the highest score of 30. However, Agile is the most suitable choice for the FixMyBD project because FixMyBD requires continuous user feedback, changing requirements, incremental delivery, modular development, rapid delivery and easy improvement after testing.

FixMyBD is a citizen-based platform where users can report issues, track progress and provide feedback. Authorities also need to update issue status and progress. Therefore, the system can benefit from regular feedback and repeated improvement. Agile supports these requirements while keeping the development process flexible and simple. The project also has an MVP, so features can be developed and delivered in small iterations.

The Spiral Model also scores highly because it supports iteration and risk analysis. However, FixMyBD is planned as a realistic prototype without direct government integration in the initial scope. Therefore, the heavy risk-analysis approach of Spiral is not necessary at this stage. Agile provides the best balance of flexibility, user feedback, incremental delivery, rapid development, and continuous improvement. Hence, Agile is selected as the most suitable SDLC model for FixMyBD.